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Gang Jiang

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Research Interests

Large Language Model, Automated Modeling, Building Energy Efficiency, Sustainability & Resilience

Education

- **University of Utah** — Salt Lake City, UT, USA
  Ph.D. in Civil Engineering — *Aug. 2022 – Jun. 2026 (Expected)*
- **Tianjin University** — Tianjin, China
  M.Sc. in Architecture & Building Sciences — *Aug. 2019 – Jun. 2022*
- **Yancheng Institute of Technology** — Yancheng, Jiangsu, China
  B.Sc. Heating, Ventilation, & Air Conditioning — *Aug. 2015 – Jun. 2019*

Research Experience

- **University of Utah, School of Computing** — Salt Lake City, UT
  Research Assistant — *May. 2025 - Present*
  - **US-NSF Projects**: (Primary Contributor) **Large language model (LLM) testing and development**: Conducted large-scale evaluation of LLMs, designed benchmarking protocols, and contributed to refining LLM architectures and training strategies for improved accuracy, efficiency, and robustness.
- **University of Utah, Civil & Environmental Engineering** — Salt Lake City, UT
  Research Assistant — *Aug. 2022 - May. 2025*
  - **US-NSF #2311685**: (Primary Contributor) Developed LLM-driven **auto-building energy modeling platforms (EPlus-LLM v1/v2, and Prompt-ABEM)** by integrating natural language with physics-based simulation.
  - **US-NSF #2318720**: (Proposal Drafting Contributor) Drafted sections on the co-simulation framework and experimental design; Conducted **co-simulations of thermo-responsive desiccants** for building dehumidification and decarbonization using EnergyPlus, Modelica, and Ansys.
  - **Global Change & Sustainability Center**: (Primary Contributor) Developed **high-fidelity building energy models** (by Resistance-Capacity mechanism and EnergyPlus software) and **efficient calibration framework** that integrates an enhanced Bayesian inference method with a deep-learning, high-resolution surrogate model, facilitating the development of digital twins for **real-time monitoring and fault detection** in real-world.
  - **US-Utah-DOT #24-8342**: (Primary Contributor) Designed a **scalable Vision-AI tool** for automated road snow condition assessment, enabling more effective snowplow operations and enhanced traffic safety.
  - **US-Utah-DOT #24-8332**: (Contributor) **Volumetric measurement of salt stockpiles using photogrammetry, LiDAR, and depth cameras**: Reconstructed 3D point cloud models of all stockpiles in CloudCompare and conducted volume calibration and results evaluation.
- **Tianjin University** — Tianjin, China
  Research Assistant — *Aug. 2019 - Jun. 2022*
  - **China-NSFs**: Developed **fault detection & diagnosis** methods for building systems with **incomplete data**, integrating Modelica simulations with machine learning to enhance accuracy and robustness.
  - **Industry Projects**: Conducted research on building energy efficiency and management, focusing on **optimizing energy efficiency and sustainability strategies**.

- **Amazon Web Services (AWS)** Beijing, China
  Intern *Jun. 2021 - Dec. 2021*
  - **Data Center Design**: Assisted in **planning and optimizing** data center infrastructure, including local generators, uninterruptible power supplies (UPS), power distribution, cooling systems, and network architecture to enhance **resilience and scalability**.
  - **Data Center Operation**: Contributed to improving **energy efficiency and fault detection** in data centers through operational optimizations and system monitoring .

Journal Publications

[7] G. Jiang, J. Chen. Efficient Fine-Tuning of Large Language Models for Automated Building Energy Modeling in Complex Cases. *Automation in Construction*, Jul. 2025.

[6] Z. Ma, G. Jiang, J. Chen. A Neural Ordinary Differential Equations-Based Approach for Enhanced Building Energy Modeling on Small Datasets. *Building Simulation*, Apr. 2025.

[5] Z. Ma, G. Jiang, Y. Hu, J. Chen. A Review of Physics-Informed Machine Learning for Building Energy Modeling. *Applied Energy*, Mar. 2025.

[4] G. Jiang, Z. Ma, L. Zhang, J. Chen. Prompt Engineering to Inform Large Language Models in Automated Building Energy Modeling. *Energy*, Feb. 2025.

[3] Z. Ma, G. Jiang, J. Chen. Physics-Informed Ensemble Learning with Residual Modeling for Enhanced Building Energy Prediction. *Energy and Buildings*, Nov. 2024.

[2] G. Jiang, Y. Chen, Z. Wang, K. Powell, B. Billings, J. Chen. A Deep Learning-Based Bayesian Framework for High-Resolution Calibration of Building Energy Models. *Energy and Buildings*, Nov. 2024.

[1] G. Jiang, Z. Ma, L. Zhang, J. Chen. EPlus-LLM: A Large Language Model-Based Computing Platform for Automated Building Energy Modeling. *Applied Energy*, Aug. 2024. Highly Cited

Project Reports

[3] M. Abbas, M. Pouria, K. Biao, G. Jiang, R. Abbas, J. Chen, W. Jessica. Volumetric Measurement of Salt Piles Using Photogrammetry, LiDAR, and Depth Cameras. *Utah Department of Transportation*, Apr. 2025.

[2] G. Jiang, B. Kuang, Z. Shi, Z. Ma, J. Chen. Snow Plow Performance Measures in Non-RWIS Locations. *Utah Department of Transportation*, Aug. 2024.

[1] G. Jiang, J. Chen, P. Kody, B. Blake. High-Fidelity Building Modeling Enabled Digital Twin Development for Continuous Building and HVAC Monitoring. *SEED2SOIL Programe, Global Change Sustainability Center*, Aug. 2023.

Conference Publications

[3] G. Jiang, J. Chen. Automated Building Energy Modeling Using Large Language Model. *COBEE*, Jul. 2025.

[2] G. Jiang, J. Chen. Fine-Tuning Large Language Models for Automated Building Simulation. *BAS*, Jul. 2025.

[1] G. Jiang, L. Zhang, J. Chen. EPlus-LLM: A Novel Automated Building Simulation Platform Using

Natural Language Processing. *ASHRAE Annual Conference*, Jul. 2024.

Teaching Experience

- **University of Utah** Salt Lake City, UT
  A Course in Green Building & Building Energy Modeling *Aug. 2024 - Dec. 2024*
  - **Computer Lab Instructor** : Led a 3-week hands-on session on building energy modeling using EnergyPlus & OpenStudio, focusing on sustainable building design, retrofit, and energy efficiency.
  - **Teaching Assistant** : Assisted with computer lab sessions, supporting course logistics, preparing class materials, slices, and syllabus, and helping students with hands-on coding assignments.

Conference & Workshop Presentations

- **ASHRAE Winter**: Transformation: Democratizing Building Energy Modeling through Large Language Models. Feb. 2026. Invited Oral Presentation
- **BuildNext**: Toward Automated Building Energy Modeling with Large Language Models. Oct. 2025. Invited Oral Presentation
- **ASHRAE CIDCO**: Automating Building Energy Modeling from Natural Language. Aug. 2025. Invited Oral Presentation
- **ASHRAE Annual**: EPlus-LLM: A Novel Automated Building Simulation Platform Using Natural Language. Jun. 2024. Oral Presentation
- **GCSC Environment and Sustainability Research Symposium**: Improved Bayesian Calibration of Campus Building Energy Models. Feb. 2023. Poster

Services

- **GitHub Maintainer**: I maintain the open-source GitHub repositories EPlus-LLM and Prompt-ABEM, and actively respond to issues.
- **Independent Reviewer**: *Renewable & Sustainable Energy Reviews*; *Cell Reports Physical Science*
- **Associate Member**: ASHRAE

Programming Skills

- **Programming Languages**: Python, PyTorch, JavaScript
- **Technologies**: High-Performance Computing on Linux, Model Deployment, GUI Development
- **Software**: EnergyPlus, SketchUp, Matlab, Dymola (Modelica)